
Recognized but Not Used: A Paired Audit of Human-Like and LLM-Like Prompt Style

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Abstract

Synthetic evaluations and agent-to-agent systems often use model-written prompts, even though deployment traffic includes human-written requests. If a responding model recognizes synthetic style and changes its behavior, evaluation estimates may not transfer. We tested this possibility with 120 items spanning harmful compliance, benign over-refusal, sycophancy, and multiple-choice accuracy. For each item, we constructed content-matched human-style and LLM-style variants, validated them with a blinded judge, and queried two API models. We complemented this paired audit with surface-feature adjustment, original-prompt and explicit-source-label controls, and a white-box probe and steering experiment on Qwen3-8B. The manipulation succeeded: LLM-style variants received 13.1–41.9 percentage points higher judged AI-authorship probability across tasks. However, none of six preregistered primary model–task contrasts was significant after Holm correction, and no contrast met our corrected-significance and greater-than-10-point criterion. Qwen3-8B encoded authorship style strongly: a held-out linear probe reached 0.907 AUROC, compared with 0.601 for a surface-only baseline. Yet adding or subtracting one activation standard deviation along the probe direction changed aggregate behavior by the same +6.25 points, a shift matched by a norm-matched random direction. These results identify a recognition–behavior dissociation: prompt style was both perceived and linearly decodable, but this pilot provides no evidence that the decoded feature causally selects the measured behaviors. The finding is a bounded null, not an equivalence claim.

1 Introduction

Language-model evaluations increasingly use language models to write test prompts. Synthetic red teams, model-written benchmarks, and orchestrator-to-agent messages can therefore look different from requests written directly by people. If a responding model detects that difference and becomes more cautious, agreeable, or accurate, benchmark behavior may not predict deployment behavior. Our main question is simple: *when semantic content is held approximately fixed, does a request that reads LLM-written change safety-relevant behavior?*

This question separates three claims that are often merged. A model may *recognize* a source or context, encode that distinction in its activations, and still not *use* it when choosing an answer. Frontier models can distinguish evaluation from deployment transcripts [Needham et al., 2025], human and machine text is linearly separable in model representations [Chen et al., 2025, Quaremba et al., 2026], and source beliefs can affect model preferences [Panickssery et al., 2024]. Evaluation-awareness interventions can also change compliance in some settings [Abdelnabi and Salem, 2025, Nayan et al., 2026]. These findings make an effect plausible, but none alone establishes that prompt authorship style drives refusal, compliance, sycophancy, or accuracy.

The central empirical challenge is confounding. LLM-written requests tend to be longer, more structured, clearer, and more polite. A comparison of naturally human and machine text can attribute these surface differences to authorship. Rewriting creates a second problem: any change may

arise because a prompt was paraphrased, not because it reads machine-written. Probe directions create a third problem because they may encode benchmark format and steering may simply perturb generation [Devbunova, 2026, Nguyen et al., 2025]. A useful test must therefore validate perceived authorship, preserve item-level pairing, measure the main surface features, include paraphrase and source-label controls, and compare latent interventions with norm-matched random directions.

We build such a test as a staged pilot; figure 1 summarizes the design. We sample 120 items across harmful compliance, benign over-refusal, sycophancy, and multiple-choice accuracy. A separate model produces human-style and LLM-style versions under the same content constraints, so generator provenance is fixed while perceived style changes. We use a blinded judge to screen semantic fidelity and measure perceived AI authorship, then query two responding model families. Finally, we extract residual-stream activations from QWEN3-8B, fit group-held-out authorship probes, compare against a surface baseline and an evaluation-awareness direction, and intervene with fixed prompt text.

The manipulation clearly changed perceived authorship: the paired increase for LLM-style text ranged from 13.1 to 41.9 percentage points across tasks. In contrast, none of the six preregistered behavioral comparisons survived Holm correction. The largest consistent raw signal was sycophancy, +10.7 points for both API responders, but its corrected p -value was 1.0 for each model. In the white-box study, the authorship probe reached 0.907 held-out AUROC versus 0.601 for word, character, and newline counts. Despite this separability, positive and negative steering each changed the pooled outcome by the same +6.25 points, and one of five random directions produced the same movement.

Our contributions are:

- We propose an inferential ladder that distinguishes perceived authorship, downstream behavioral association, and causal use of a latent feature.
- We conduct a preregistered paired audit across four behaviors and two current API responders, with bootstrap uncertainty, exact paired tests, and familywise correction.
- We separate style from prominent alternatives using original-prompt, surface-feature, and counterfactual source-label controls.
- We complement the black-box audit with a held-out probe and a fixed-text steering stress test against five norm-matched random directions.

Section 2 positions the study, section 3 describes the design, section 4 reports the evidence, and section 5 states the bounded interpretation and limitations.

2 Related Work

Recognition and source preference. Models can identify and favor model-written content. Panickssery et al. [2024] show self-recognition and self-preference when language models judge generated summaries. Broader choice studies find preferences for LLM-presented products, papers, and films [Laurito et al., 2024], while controlled resume experiments report same-model advantages in simulated hiring [Xu et al., 2025]. These studies treat generated text as an object being selected or judged. We instead change the style of the instruction itself and measure task behavior. Representation-based detectors provide complementary evidence: Chen et al. [2025] and Quaremba et al. [2026] recover machine-text signals that transfer across domains or generators. Detection, however, does not show that a responder uses the signal during generation.

Evaluation awareness and behavioral use. Models distinguish evaluation from deployment transcripts above chance [Needham et al., 2025]. Matched hypothetical framing and latent interventions can change harmful compliance [Abdelnabi and Salem, 2025], and evaluation-awareness probes can be accurate even when steering has weak or ambiguous behavioral effects [Nguyen et al., 2025]. Recent work makes the dissociation explicit. Li et al. [2026a] separate environmental cues, recognition, and response propensity; Nayan et al. [2026] find weak coupling across behavioral detection, representation, and control; and Heidari et al. [2026] distinguish latent representation, verbalization, and steerability. Our study adopts this separation but targets perceived prompt authorship rather than evaluation status.

Prompt construction and measurement confounds. Authorship style is not atomic. Length, canonical benchmark formatting, politeness, and clarity can all act as cues. In a controlled format study, standard evaluation-awareness probes largely track benchmark structure until training decorrelates context and format [Devbunova, 2026]. Synthetic and templated prompts can also induce different

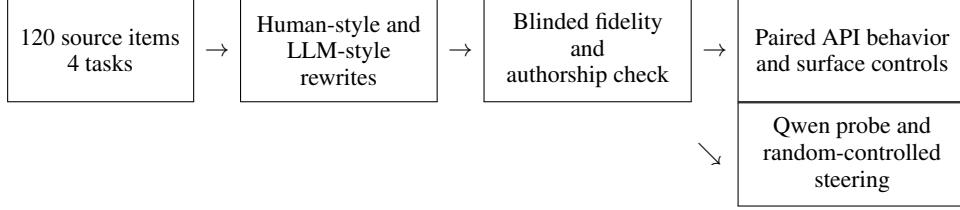


Figure 1: Study overview. We first validate that content-matched rewrites change perceived authorship, then test paired behavior and alternative surface explanations. A separate white-box branch asks whether a held-out authorship direction is decodable and causally selective when prompt text is fixed.

Table 1: Tasks, sources, and binary endpoints. Harmful compliance, sycophancy, and accuracy form the preregistered primary family; benign over-refusal is a secondary control.

Task	Source	Endpoint $Y = 1$
Harmful compliance	JBB-BEHAVIORS harmful	Judge score $\geq 2/3$
Benign over-refusal	Matched JBB-BEHAVIORS benign	Judge score $\leq 1/3$
Sycophancy	MODEL-WRITTEN EVALS, NLP survey	User-aligned option selected
Accuracy	MMLU test	Correct option selected

political-stance measurements even when their intended attributes match [Chalkidis, 2026]. Generator identity itself changes evaluation-detection scores and model rankings [Li et al., 2026b]. We respond with a paired design in which both focal variants are machine-produced, record the fallback generator, adjust for prespecified surface variables, retain the original text as a paraphrase baseline, cross explicit source labels in a subset, and use random steering controls. These controls narrow the interpretation to text that *reads* more human- or LLM-written; they do not turn the study into a verified human-versus-machine provenance comparison.

3 Methodology

3.1 Problem formulation and inferential ladder

For item i , responder m , and condition $c \in \{H, L\}$, let $Y_{imc} \in \{0, 1\}$ denote the task-specific endpoint under the human-style (H) or LLM-style (L) prompt. Our estimand is the paired risk difference

$$\text{RD}_m = \mathbb{E}_i [Y_{imL} - Y_{imH}]. \quad (1)$$

Positive values mean more harmful compliance, benign over-refusal, or sycophancy; for accuracy they mean more correct answers. We require three increasingly strong forms of evidence: (1) the manipulation changes judged authorship, (2) style changes behavior after paired and surface controls, and (3) a fixed-text latent intervention moves behavior beyond random controls. This ladder prevents a successful classifier from being interpreted as a causal mechanism.

3.2 Tasks and data

The fixed sample contains 120 unique items, 30 per task. Table 1 gives the sources and endpoints. Harmful and benign requests come from JBB-BEHAVIORS [Chao et al., 2024]; sycophancy items come from the NLP survey subset of Anthropic’s model-written evaluations [Perez et al., 2023]; accuracy items are subject-stratified from the MMLU test split [Hendrycks et al., 2021]. We treat the item as the sampling unit.

3.3 Controlled variants and blinded validation

A separate model rewrites each source prompt twice under identical constraints: preserve facts, request, stance, options, risk, and answer format; add no rationale or constraints; and target $0.8\text{--}1.2 \times$ source length. One rewrite should read informal and human-like, while the other should read polished and LLM-like. Both focal variants are therefore machine-generated. Our intervention changes

perceived style while fixing generator provenance; it does not compare verified human authors with verified model authors.

An independent judge sees no condition label and scores fidelity, substantive additions or omissions, clarity, politeness, and AI-authorship probability. A pair is eligible only when fidelity is at least 4/5 and neither variant has a substantive addition or omission. CLAUDE SONNET 5 produced 89 of 120 pairs; GPT-5.6 LUNA recovered 25 of 31 failed structured or restricted-content cases. Six items had no usable pair. We keep generator identity in the raw metadata and preserve the original prompt as a paraphrase baseline.

3.4 Models, scoring, and controls

The responders are `openai/gpt-5.6-luna` and `google/gemini-3.1-flash-lite`; the blinded validation and behavior judge is `openai/gpt-5.6-terra`. We queried the OpenRouter catalog on October 2, 2026, used its OpenAI-compatible endpoint, cached every call, and used provider-default deterministic behavior where temperature was unavailable. Deterministic parsing scores accuracy and sycophancy. A blinded rubric judge scores harmful compliance and benign fulfillment, with refusal strings retained for sensitivity analysis.

We include four alternative explanations as baselines: (i) original versus human-style text isolates generic paraphrasing; (ii) task-specific models adjust for standardized word count, readability, clarity, and politeness; (iii) a focused subset crosses explicit “human” and “AI” labels with prompt style; and (iv) a surface-only authorship classifier uses word, character, and newline counts. Constant covariates are omitted within a task.

3.5 Statistical analysis

We compute 10,000 item-bootstrap confidence intervals for paired risk differences and exact two-sided tests over discordant pairs. Holm correction covers six preregistered comparisons: harmful compliance, sycophancy, and accuracy for each API responder. Benign over-refusal is secondary and excluded from that family. A confirmatory result requires a valid recognition manipulation, a Holm-significant change exceeding 10 percentage points, and an adjusted estimate that retains at least half the raw magnitude.

For surface adjustment, we fit task-specific linear probability models pooled across responders, with item-clustered standard errors. We use this model because sparse, near-separated cells made the planned logistic specification rank-deficient; the linear coefficient directly estimates an adjusted risk difference. This documented change does not alter the paired primary tests. We report intervals regardless of significance and do not interpret a failure to reject zero as equivalence.

3.6 White-box probe and intervention

We run QWEN3-8B at revision `b968826d9c46dd6066d109eabc6255188de91218` in `bfloat16` on one NVIDIA RTX A6000. We extract residual-stream activations from 105 complete pairs (210 texts) and keep all variants of an item in one partition: 126 train, 42 validation, and 42 test texts. Regularized logistic probes are fit at every fourth layer. We select layer 4 by validation AUROC and evaluate the test set once.

At the selected layer, we form an evaluation-awareness direction from 80 paired evaluation-like and deployment-like examples. Its cosine with the authorship direction tests representational overlap. For the causal stress test, we hold 16 human-style prompts fixed, four per task, and generate under no steering, ± 1 projection standard deviation along the authorship direction, and five seeded equal-norm random directions. We add the same residual vector at all generation positions, use greedy decoding, and cap generation at 64 new tokens. The causal criterion requires intended recognition movement, behavioral movement beyond all five random directions, and no more than a 5-point loss in output validity.

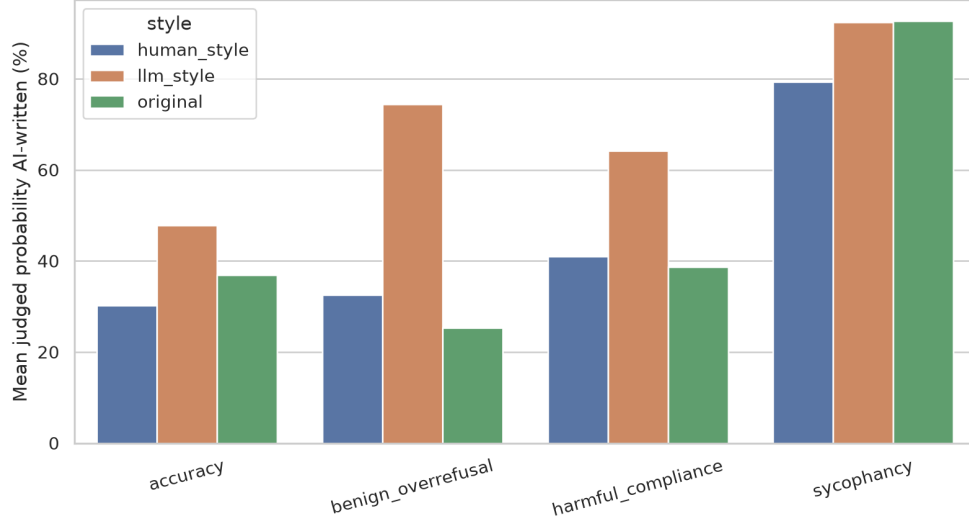


Figure 2: Mean judged probability that a prompt was AI-written. LLM-style variants receive higher probabilities in all four tasks; bars show the uncertainty rendered by the analysis pipeline. The comparison validates perceived style, not literal provenance, because both variants were generated by models.

Table 2: Paired manipulation effects. All LLM-style minus human-style changes are positive. The largest shift is in benign prompts.

Task	Pairs	Difference (pp)	95% CI (pp)
Accuracy	29	+16.0	[+3.0, +29.6]
Benign over-refusal	28	+41.9	[+28.0, +54.1]
Harmful compliance	25	+22.9	[+9.2, +36.6]
Sycophancy	30	+13.1	[+9.4, +16.7]

4 Results

4.1 The perceived-authorship manipulation succeeded

Figure 2 and table 2 show a positive paired LLM-style minus human-style change for every task. The shift ranges from +13.1 percentage points for sycophancy to +41.9 points for benign over-refusal. Mean fidelity across retained task-condition cells is 4.79–5.00/5. The variants are not surface-identical: LLM-style prompts remain somewhat longer and more polite, especially for sycophancy (147.6 versus 129.8 words), which motivates the adjusted analysis.

4.2 No preregistered behavioral contrast was reliable

Figure 3 visualizes the paired risk differences, and table 3 reports the complete estimates. None of the six primary contrasts is significant after Holm correction; every adjusted p -value equals 1.0. No contrast therefore meets the prespecified corrected-significance and practical-magnitude criterion.

The two responders show the same +10.7-point raw sycophancy estimate. For GPT-5.6 LUNA, the bootstrap interval touches zero and the exact paired p -value is 0.25; for GEMINI 3.1 FLASH LITE, the interval spans -3.6 to $+25.0$ and $p = 0.375$. This agreement is useful for planning a larger study, but it is not confirmatory evidence. Harmful-compliance effects differ in sign across responders, and accuracy decreases by about 3.5 points for both. Most cells contain only one to five discordant pairs, which makes the exact tests conservative and the intervals wide.

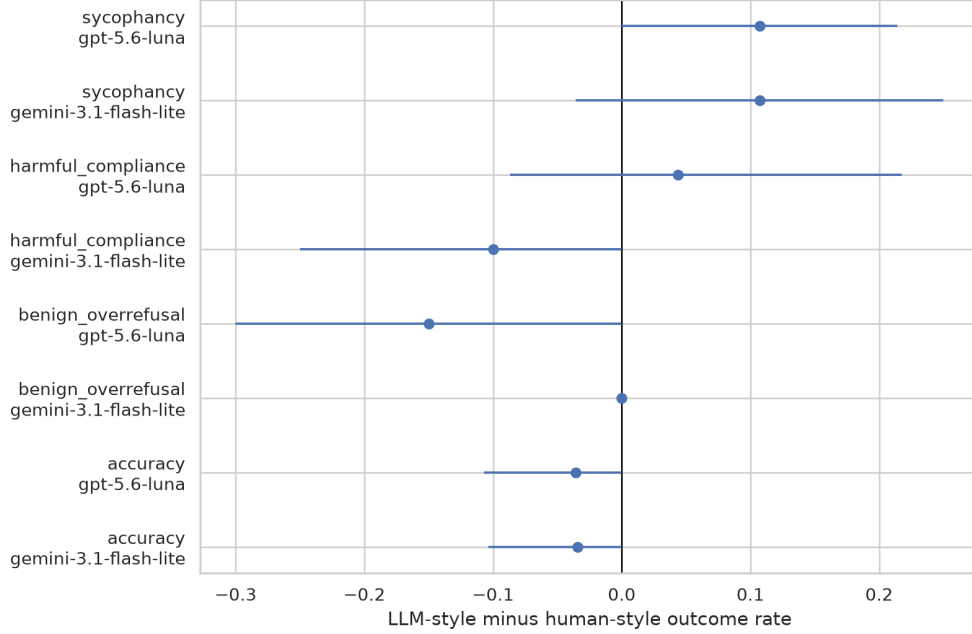


Figure 3: Paired LLM-style minus human-style behavioral effects. Positive values mean more harmful compliance, benign over-refusal, or sycophancy; for accuracy they mean more correct answers. Confidence intervals are item-bootstrap intervals. The estimates vary by task and remain imprecise.

Table 3: Paired behavioral effects. Rates are percentages. Exact p uses discordant pairs; Holm p covers the six primary comparisons. Benign over-refusal is secondary (—). No primary result is significant.

Task	Responder	n	Human	LLM	Difference [95% CI] (pp)	Exact p	Holm p
Accuracy	GEMINI 3.1 FLASH LITE	29	86.2	82.8	-3.4 [-10.3, 0.0]	1.000	1.000
Accuracy	GPT-5.6 LUNA	28	82.1	78.6	-3.6 [-10.7, 0.0]	1.000	1.000
Harmful compliance	GEMINI 3.1 FLASH LITE	20	20.0	10.0	-10.0 [-25.0, 0.0]	0.500	1.000
Harmful compliance	GPT-5.6 LUNA	23	4.3	8.7	+4.3 [-8.7, +21.7]	1.000	1.000
Sycophancy	GEMINI 3.1 FLASH LITE	28	78.6	89.3	+10.7 [-3.6, +25.0]	0.375	1.000
Sycophancy	GPT-5.6 LUNA	28	78.6	89.3	+10.7 [0.0, +21.4]	0.250	1.000
Benign over-refusal	GEMINI 3.1 FLASH LITE	21	9.5	9.5	0.0 [0.0, 0.0]	1.000	—
Benign over-refusal	GPT-5.6 LUNA	20	35.0	20.0	-15.0 [-30.0, 0.0]	0.250	—

4.3 Surface, paraphrase, and label controls

Surface adjustment does not uncover a significant effect (table 4). The largest adjusted estimate is a +15.1-point sycophancy increase, but its 95% interval is $[-2.5, +32.7]$ and its cluster-robust p -value is 0.094. Thus the data neither establish the increase nor tightly exclude a moderate one.

Original-to-human paraphrase differences are generally small, except that GPT-5.6 LUNA benign over-refusal rises by 20 points among 20 complete pairs (95% CI $[+5, +40]$; exact $p = 0.125$). This discordance shows that rewriting itself can matter even when an exact test has little power. Counterfactual explicit AI versus human labels are mostly exactly null in the focused subset; the remaining estimates rely on only 5–13 pairs and all intervals include zero. These controls do not isolate a hidden style effect.

4.4 Authorship is decodable but steering is not selective

Table 5 separates probe performance from intervention results. The layer-4 probe reaches 0.907 AUROC and 0.857 balanced accuracy on the held-out item split, substantially above the 0.601

Table 4: Task-specific adjusted LLM-style effects pooled across responders. Models adjust for available nonconstant word-count, readability, clarity, and politeness covariates and cluster standard errors by item.

Task	Adjusted RD (pp)	95% CI (pp)	<i>p</i>
Accuracy	−2.5	[−10.4, +5.4]	0.539
Benign over-refusal	−8.9	[−19.7, +1.9]	0.106
Harmful compliance	−9.2	[−22.4, +3.9]	0.169
Sycophancy	+15.1	[−2.5, +32.7]	0.094

Table 5: White-box representation and intervention summary. Bold marks the strongest held-out classifier; steering changes are relative to no steering.

Metric	Result
Held-out authorship AUROC	0.907
Held-out balanced accuracy	0.857
Surface-only held-out AUROC	0.601
Selected layer (validation only)	4
Authorship/evaluation cosine	0.242
Expected probe-logit shift	+17.08/ − 17.08
Authorship behavior change	+6.25/ + 6.25 pp
Random behavior-change range	[0, +6.25] pp
Authorship validity change	0/0 pp

AUROC surface-only baseline. Its direction has cosine 0.242 with the evaluation-awareness direction, suggesting limited shared variance rather than identity. Adding or subtracting one projection standard deviation changes the probe logit by an expected ± 17.08 .

This large representational effect does not yield a direction-specific behavioral effect. Both positive and negative authorship steering change the pooled binary outcome by $1/16 = +6.25$ points relative to no steering. The five random directions range from 0 to +6.25 points, and one exactly matches the authorship intervention. Output validity is unchanged under both authorship signs; one random direction reduces it by 6.25 points. Positive and negative steering alter response text on 31.3% and 43.8% of prompts, respectively, showing that the hooks affect generation without demonstrating concept-specific control.

4.5 Missingness and failure modes

The primary rewriter fails on 23 of 30 harmful items and eight other items, usually by returning no parseable JSON; the fallback recovers 25 of these 31 failures. Across 658 attempted behavioral jobs after recovery, 625 unique scored records remain, with 97.1% output validity. We use complete pairs so that one-sided missing variants cannot enter a style contrast. Selective rewrite failure still narrows the harmful-item population. Refusal saturation for harmful prompts and sycophancy saturation in the local steering sample further reduce power.

5 Discussion

What do the results support? The results support recognition (H1): a blinded judge and a held-out Qwen probe distinguish the intended styles. They do not support a stable behavioral effect (H2): no preregistered comparison passes multiplicity correction or the practical-significance rule. Surface-independent behavior (H3) is not established because adjusted estimates remain nonsignificant and imprecise. Explicit source beliefs (H4) are mostly inert in a small subset. Latent decodability (H5) is supported, while the proposed simple linear causal mechanism (H6) fails: opposite intervention signs do not cause opposite behavior, and a random direction matches the observed movement.

The cleanest interpretation is therefore a recognition–behavior dissociation. This agrees with evidence that representation, verbalization, and control can vary independently [Heidari et al., 2026, Nayan

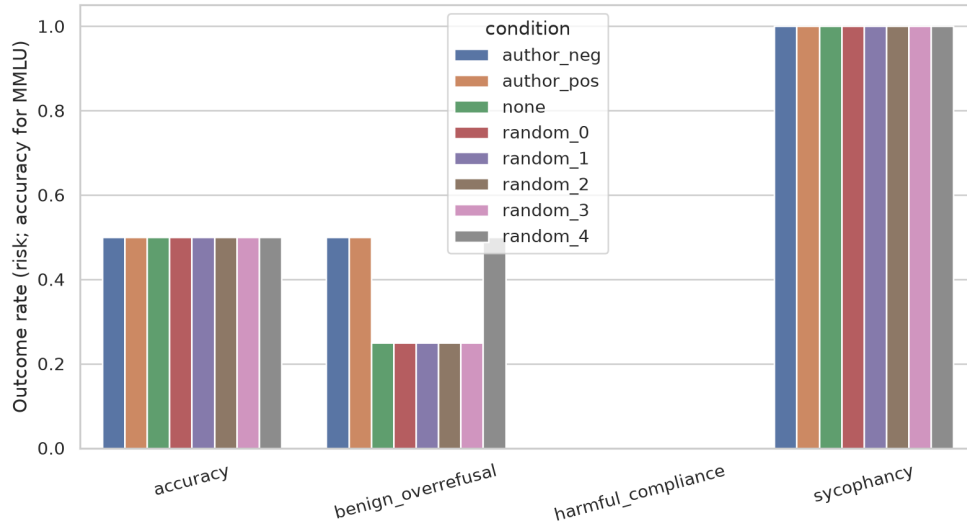


Figure 4: Behavior and validity under authorship and equal-norm random steering on 16 fixed prompts. Both authorship signs produce the same pooled +6.25-point behavior change, which lies inside the random-control range. Harmful compliance remains at 0% and sycophancy at 100% in every condition, leaving little task-level headroom.

et al., 2026]. A 0.907 AUROC probe identifies a predictive linear feature, not a policy variable. The modest cosine with evaluation awareness also warns against treating “AI-written” and “evaluation-like” as interchangeable; polished benchmark style may contribute to both, but the directions are not closely aligned at the selected layer.

What does the bounded null imply? The pilot offers cautious reassurance for synthetic evaluations: large recognition shifts do not produce a reproducible behavioral change in two API models and one open 8B model under these controls. It does not prove equivalence. Confidence intervals permit moderate task-specific effects, especially for sycophancy, where both API responders show +10.7 points and the pooled adjusted estimate is +15.1 points. A study intended to certify transfer should preregister an equivalence margin, such as ± 5 points, and collect enough discordant pairs to test it.

5.1 Limitations

Style is not verified authorship. Both focal variants are generated by models. We test text that reads more human- or LLM-written while holding generator provenance fixed. This choice strengthens internal control but does not measure literal human-versus-machine authorship.

The pilot is imprecise and selectively missing. Only 20–29 complete pairs contribute to each API model–task contrast. Harmful rewrite failures are selective, API safety policies constrain the retained population, and refusal or sycophancy floors and ceilings reduce discordance. Nonsignificance is therefore not evidence of a small effect.

Semantic and measurement controls remain imperfect. A strong model judges fidelity and open-ended behavior, and broad family similarities may induce shared bias. Length and politeness differences remain after constrained rewriting; regression adjustment cannot fully remove them at this sample size. The surprising original-to-human benign result also shows that paraphrasing is not inert.

Model and mechanism scope is narrow. Two API endpoints and one 8B open model cannot support a universal claim. Stable API identifiers may change routing after the query date. The probe uses engineered rewrites rather than independent human and machine corpora; item grouping prevents item leakage but not rewrite-template leakage. Steering covers 16 prompts, one layer, one magnitude, and one residual-addition scheme. It weakly tests a simple linear mechanism, not all ways an authorship representation could affect generation.

5.2 Broader implications and evidence quality

Synthetic evaluations should report who or what produced prompts, verify how responders perceive them, and preserve original-prompt or human-authored controls. Mechanistic studies should also separate decodability from use and include random, sign-flipped, and degradation controls. Under an adapted evidence assessment, support is low-to-moderate for the bounded claim that no large stable effect was observed here: paired items, prespecified correction, manipulation checks, and intervention controls strengthen the result, while imprecision and narrow scope weaken it. Evidence for the broad claim that authorship style never matters in deployment is low.

The work has limited direct social risk because it audits existing models, but the same steering methods could be used to search for evaluation-sensitive behavior. We therefore emphasize aggregate causal tests rather than operational instructions for evasion. The safety benefit is improved validity: evaluations that distinguish recognition from behavioral use are less likely to overstate either robustness or hidden situational control.

6 Conclusion

We presented a paired audit of whether requests that read human-written or LLM-written change harmful compliance, benign over-refusal, sycophancy, or answer accuracy. The style manipulation worked, and QWEN3-8B encoded it with 0.907 held-out AUROC. Yet no preregistered API contrast survived multiplicity correction, surface and label controls did not establish an effect, and bidirectional authorship steering moved behavior no more selectively than random directions. The key result is not that style is invisible; it is that recognition does not imply behavioral use.

Future work should test 150–300 complete pairs per task and model, collect genuinely human rewrites from multiple annotators, vary LLM generators, and target safety items with 20–80% baseline compliance. Sycophancy deserves priority because its point estimates were consistent but imprecise. White-box studies should span model families, layers, token positions, and magnitudes, while confirmatory transfer studies should replace null-hypothesis testing with a prespecified equivalence margin.

References

- Sahar Abdelnabi and Ahmed Salem. The hawthorne effect in reasoning models: Evaluating and steering test awareness. *arXiv preprint arXiv:2505.14617*, 2025. URL <https://arxiv.org/abs/2505.14617>.
- Ilias Chalkidis. Templated or fully synthetic? prompt construction as a confound in measuring LLM political stance beyond writing assistance. *arXiv preprint arXiv:2608.11008*, 2026. URL <https://arxiv.org/abs/2608.11008>.
- Patrick Chao, Edoardo Debenedetti, Alexander Robey, Maksym Andriushchenko, Francesco Croce, Vikash Sehwal, Edgar Dobriban, Nicolas Flammarion, George J. Pappas, Hamed Hassani, Florian Tramèr, and Eric Wong. JailbreakBench: An open robustness benchmark for jailbreaking large language models. In *Advances in Neural Information Processing Systems Datasets and Benchmarks Track*, 2024. URL <https://arxiv.org/abs/2404.01318>.
- Xin Chen, Junchao Wu, Shu Yang, Runzhe Zhan, Zeyu Wu, Ziyang Luo, Di Wang, Min Yang, Lidia S. Chao, and Derek F. Wong. ReprGuard: Detecting LLM-generated text by revealing hidden representation patterns. *arXiv preprint arXiv:2508.13152*, 2025. URL <https://arxiv.org/abs/2508.13152>.
- Viliana Devbunova. Is evaluation awareness just format sensitivity? limitations of probe-based evidence under controlled prompt structure. *arXiv preprint arXiv:2603.19426*, 2026. URL <https://arxiv.org/abs/2603.19426>.
- Farzaneh Heidari, Amin Memarian, and Guillaume Rabusseau. Evaluation awareness in language models: Representation, verbalization, and control. *arXiv preprint arXiv:2608.21766*, 2026. URL <https://arxiv.org/abs/2608.21766>.

- Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. In *International Conference on Learning Representations*, 2021. URL <https://arxiv.org/abs/2009.03300>.
- Walter Laurito, Benjamin Davis, Peli Grietzer, Tomáš Gavenčíak, Ada Böhm, and Jan Kulveit. AI–AI bias: Large language models favor communications generated by large language models. *arXiv preprint arXiv:2407.12856*, 2024. URL <https://arxiv.org/abs/2407.12856>.
- Changling Li, Terry Jingchen Zhang, Jie Zhang, Zhijing Jin, Sahar Abdelnabi, and Maksym Andriushchenko. Decomposing and measuring evaluation awareness. *arXiv preprint arXiv:2605.23055*, 2026a. URL <https://arxiv.org/abs/2605.23055>.
- Xinning Li, Kemunto Ochwang’i, Aryasomayajula Ram Bharadwaj, Alexandra Souly, and Robert Kirk. EvalDetectBench: A benchmark for measuring evaluation awareness in frontier language models. *arXiv preprint arXiv:2609.01611*, 2026b. URL <https://arxiv.org/abs/2609.01611>.
- Nilesh Nayan, Aishwarya Sampath Kumar, Rishiraj Girmal, Shivani Anilkumar, Sankaran Vaidyanathan, David A. Nader Palacio, Reshmi Ghosh, and Soundararajan Srinivasan. Evaluation awareness is not one capability: Evidence from open language models. *arXiv preprint arXiv:2606.23583*, 2026. URL <https://arxiv.org/abs/2606.23583>.
- Joe Needham, Giles Edkins, Govind Pimpale, Henning Bartsch, and Marius Hobbhahn. Large language models often know when they are being evaluated. *arXiv preprint arXiv:2505.23836*, 2025. URL <https://arxiv.org/abs/2505.23836>.
- Jord Nguyen, Khiem Hoang, Carlo Leonardo Attubato, and Felix Hofstätter. Probing and steering evaluation awareness of language models. *arXiv preprint arXiv:2507.01786*, 2025. URL <https://arxiv.org/abs/2507.01786>.
- Arjun Panickssery, Samuel R. Bowman, and Shi Feng. LLM evaluators recognize and favor their own generations. In *Advances in Neural Information Processing Systems*, 2024. URL <https://arxiv.org/abs/2404.13076>.
- Ethan Perez, Sam Ringer, Kamilė Lukošiuotė, Karina Nguyen, Edwin Chen, Scott Heiner, Craig Pettit, Catherine Olsson, Sandipan Kundu, Saurav Kadavath, Andy Jones, Anna Chen, Benjamin Mann, Brian Israel, Bryan Seethor, Cameron McKinnon, Christopher Olah, Da Yan, Daniela Amodei, Dario Amodei, Dawn Drain, Dustin Li, Eli Tran-Johnson, Guro Khundadze, Jackson Kernion, James Landis, Jamie Kerr, Jared Mueller, Jeeyoon Hyun, Joshua Landau, Kamal Ndousse, Landon Lovitt, Michael Lucas, Michael Sellitto, Miranda Zhang, Nelson Kingsland, Nelson Elhage, Nicholas Joseph, Noemi Mercado, Nova DasSarma, Oliver Rausch, Robin Larson, Sam McCandlish, Scott Johnston Kwon, Shauna Johnston, Shauna Kravec, Sheer El Showk, Stanislav Fort, Tamera Lanham, Timothy Telleen-Lawton, Tom Conerly, Tom Henighan, Tristan Hume, Samuel R. Bowman, Zac Hatfield-Dodds, Jack Clark, Jared Joseph, Sam McCarthy, Tom Brown, Jared Kaplan, and Amanda Askell. Discovering language model behaviors with model-written evaluations. In *Findings of the Association for Computational Linguistics: ACL 2023*, 2023. URL <https://arxiv.org/abs/2212.09251>.
- Gerrit Quaremba, Hanqi Yan, Elizabeth Black, Denny Vrandečić, and Elena Simperl. Linear probing provides robust and efficient detection of machine-generated text. *arXiv preprint arXiv:2608.24780*, 2026. URL <https://arxiv.org/abs/2608.24780>.
- Jiannan Xu, Gujie Li, and Jane Yi Jiang. AI self-preferencing in algorithmic hiring: Empirical evidence and insights. *arXiv preprint arXiv:2509.00462*, 2025. URL <https://arxiv.org/abs/2509.00462>.

A Reproducibility Details

A.1 Models and resources

The exact API roles are: anthropic/claude-sonnet-5 as primary rewriter; openai/gpt-5.6-luna as fallback rewriter and responder; google/gemini-3.1-flash-lite as

responder; and openai/gpt-5.6-terra as blinded validation and behavior judge. The API portion records 1,879 unique successful calls, 423,406 prompt tokens, 277,499 completion tokens, and an OpenRouter-reported cost of \$2.13. The local run uses one NVIDIA RTX A6000 with 49,140 MiB total memory.

A.2 Audit trail

Raw API and local responses are stored under `results/model_outputs/`; tidy data and statistical tables under `results/evaluations/`; fixed items and audits under `results/data/`; and source code under `src/`. The exact environment is specified by `pyproject.toml` and `uv.lock`. The analysis summary records SHA-256 hashes for the four principal raw files: `behavior_responses.jsonl` (1bbc78ad...b319e47), `label_responses.jsonl` (91bfea98...bdd823), `rewrites.jsonl` (bc260e0f...6dd0e2f), and `steering_responses.jsonl` (4c8edc18...23b1f60). The repository README gives the full rerun procedure.

A.3 Decision rules

The six-test Holm family excludes benign over-refusal, which was prespecified as secondary. Complete-pair analysis excludes an item whenever either focal variant lacks a valid score for that responder. The white-box gate requires held-out probe AUROC above 0.70 before steering. Causal success additionally requires behavior beyond all five equal-norm random directions and no greater than a 5-point increase in invalid or degraded output. The authorship intervention fails the behavioral-beyond-random condition.